Synthesizing a RTL Design

Objectives

After completing this lab, you will be able to:

- Use the provided Xilinx Design Constraint (XDC) file to constrain the timing of the circuit
- Elaborate the design and understand the output
- Synthesize the design with the provided basic timing constraints
- Analyze the output of the synthesized design
- Change the synthesis settings and see their effects on the generated output
- Write a checkpoint after the synthesis so the results can be analyzed after re-loading it

Steps

Create a Vivado Project using IDE

In this design we will use board's USB-UART which is controlled by the Zynq's ARM Cortex-A9 processor. Our PL design needs access to this USB-UART. So first thing we will do is to create a Processing System (PS) design which will put the USB-UART connections in a simple GPIO-style and make it available to the PL section.

Launch Vivado and create a project targeting the XC7Z020clg400-1 device, and use provided the tcl scripts (ps7_create_pynq.tcl) to generate the block design for the PS subsystem. Also, add the Verilog HDL files, uart_led_pins_pynq.xdc and uart_led_timing.xdc files from the < *2018_2_zynq_sources* >*lab2* directory.

<2018_2_zynq_labs> refers to **C:\xup\fpga_flow\2018_2_zynq_labs** directory and **<2018_2_zynq_sources>** refers to **C:\xup\fpga_flow\2018_2_zynq_sources** directory.

1. Open Vivado by selecting **Start > Xilinx Design Tools > Vivado 2018.2**
2. Click **Create New Project** to start the wizard. You will see *Create A New Vivado Project* dialog box. Click **Next**.
3. Click the Browse button of the *Project location* field of the **New Project** form, browse to **<2018_2_zynq_labs>**, and click **Select**.
4. Enter **lab2** in the *Project name* field. Make sure that the *Create Project Subdirectory* box is checked. Click **Next**.
5. Select **RTL Project** option in the *Project Type* form and click **Next**.
6. Using the drop-down buttons, select **Verilog** as the *Target Language* and *Simulator Language* in the *Add Sources* form.
7. Click on the **Blue Plus** button, then the **Add Files…** button and browse to the **<2018_2_zynq_sources>\lab2** directory, select all the Verilog files *(led_ctl.v, meta_harden.v, uart_baud_gen.v, uart_led.v, uart_rx.v, uart_rx_ctl.v and uart_top.v),* click **OK**, and then click **Next** to get to the *Add Constraints* form.

8. Click on the **Blue Plus** button, then **Add Files...** and browse to the **<2018_2_zynq_sources>\lab2** directory (if necessary), select *uart_led_timing_pynq.xdc* and click **Open**.
9. Click **Next.**

   This Xilinx Design Constraints file assigns the basic timing constraints (period, input delay, and output delay) to the design.
10. In the *Default Part* form, Use the **Boards** option, you may select the **PYNQ-Z1** or the **PYNQ-Z2** depending on your board from the Display Name drop down field.

    You may also use the **Parts** option and various drop-down fields of the **Filter** section. Select the **XC7Z020clg400-1 part**.

    Note: Notice that PYNQ-Z1 and PYNQ-Z2 may not be listed under Boards menu as they are not in the tools database. If not listed then you can download the board files for the desired boards either from Digilent PYNQ-Z1 web page or TUL PYNQ-Z2 web page.
11. Click **Next**.
12. Click **Finish** to create the Vivado project.
13. In the Tcl Shell window enter the following command to change to the lab directory and hit **Enter**.

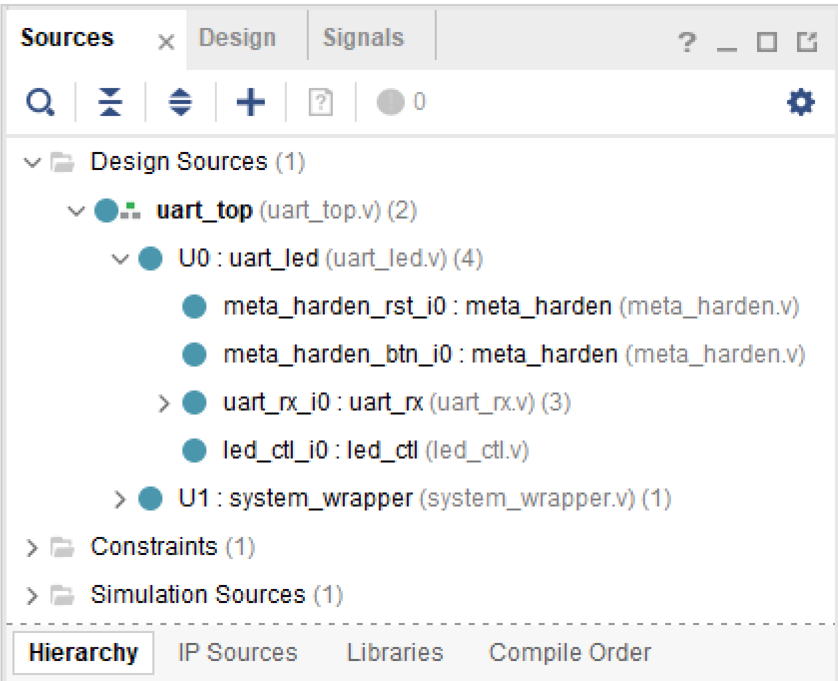
    cd C:/xup/fpga_flow/2018_2_zynq_sources/lab2
14. Generate the PS design by executing the provided Tcl script.

    source ps7_create_pynq.tcl

    This script will create a block design called *system*, instantiate ZYNQ PS with one GPIO channel (GPIO14) and one EMIO channel. It will then create a top-level wrapper file called system_wrapper.v which will instantiate the system.bd (the block design). You can check the contents of the tcl files to confirm the commands that are being run.

Analyze the design source files hierarchy.

1. In the *Sources* pane, expand the **uart_led** entry and notice hierarchy of the lower-level modules.



Opening the source file

2. Double-click on the **uart_led** entry to view its content.

   Notice in the Verilog code, the BAUD_RATE and CLOCK_RATE parameters are defined to be 115200 and 125 MHz respectively as shown in the design diagram. Also notice that the lower level modules are instantiated. The meta_harden modules are used to synchronize the asynchronous reset and push-button inputs.

```verilog

`timescale 1ns/1ps
module uart_led (
  // Write side inputs
  input              clk_pin,      // Clock input (from pin)
  input              rst_pin,      // Active HIGH reset (from pin)
  input              btn_pin,      // Button to swap high and low bits
  input              rxd_pin,      // RS232 RXD pin - directly from pin
  output       [7:0] led_pins      // 8 LED outputs
);

//***************************************************************************
// Parameter definitions
//***************************************************************************
  parameter BAUD_RATE           = 115_200;
  parameter CLOCK_RATE          = 125_000_000;

```

CLOCK_RATE parameter of uart_led for the PYNQ board

3. Expand **U0** and **uart_rx_i0** instance to see its hierarchy.

This module uses the baud rate generator and a finite state machine. The rxd_pin is sampled at a x16 the baud rate.

Open the uart_led_timing_pynq.xdc source and analyze the content

1. In the *Sources* pane, expand the *Constraints* folder and double-click the **uart_led_timing_pynq.xdc** entry to open the file in text mode.

```
1  ##pynq Timing
2
3  # define clock and period
4  create_clock -period 8.000 -name clk_pin -waveform {0.000 4.000} [get_ports clk_pin]
5
6  # Create a virtual clock for IO constraints
7  create_clock -period 12.000 -name virtual_clock
8
9  #input delay
10 set_input_delay -clock virtual_clock -max 0.000 [get_ports rst_pin]
11 set_input_delay -clock virtual_clock -min -0.500 [get_ports rst_pin]
12
13 set_input_delay -clock virtual_clock -max 0.000 [get_ports btn_pin]
14 set_input_delay -clock virtual_clock -min -0.500 [get_ports btn_pin]
15
16 #output delay
17 set_output_delay -clock virtual_clock 0.000 [get_ports {led_pins[*]}]
```

Timing constraints

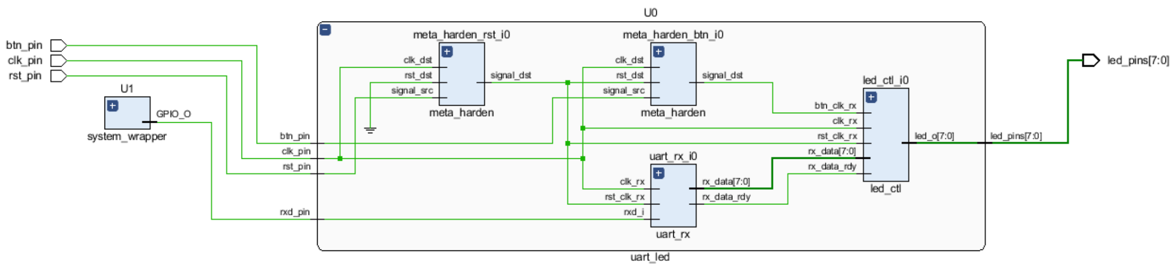
Line 4 creates the period constraint of 8ns with a duty cycle of 50%. Line 7 creates a virtual clock of 12 ns. This clock can be viewed as the upstream device is generating its output with respect to its clock and outputs data with respect to it. The btn_pin is constrained with respect to the upstream clock (lines 13, 14). The led_pins are constrained with respect to the upstream clock as the downstream device may be using it.

Elaborate the Design

Elaborate and perform the RTL analysis on the source file.

1. Click **Flow Navigator > RTL ANALYSIS > Open Elaborated Design > Schematic**. Click **OK**.

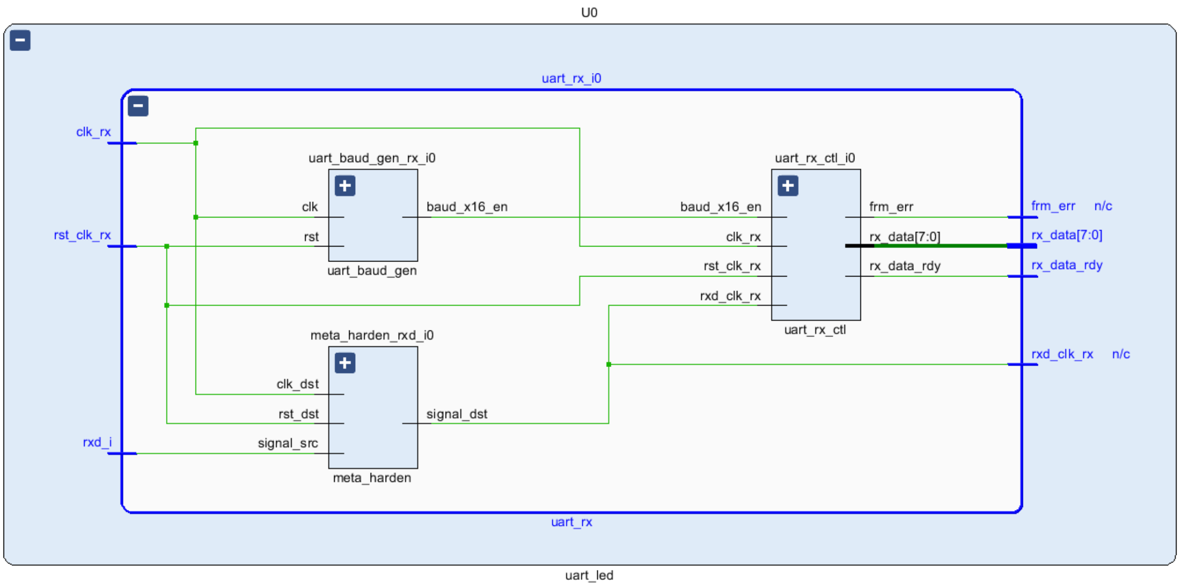
   The model (design) will be elaborated and a logical view of the design is displayed.



A logic view of the design one-level down from the top in component U0

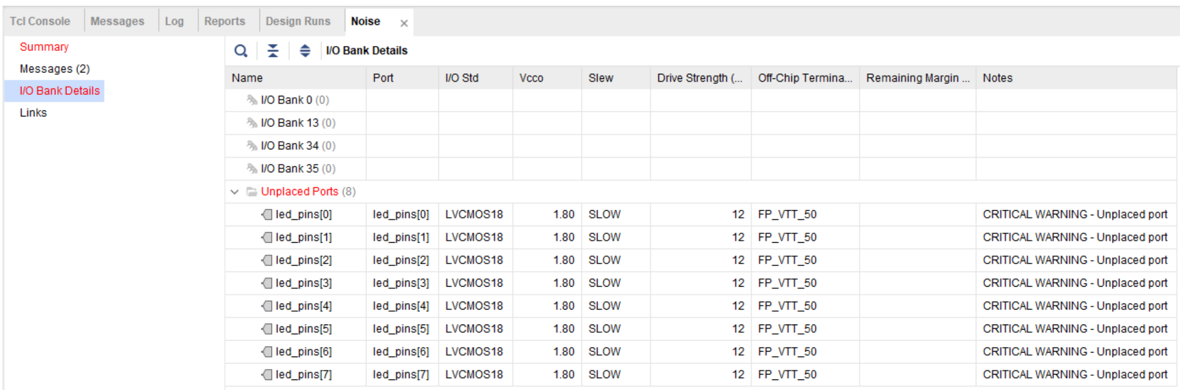
You will see two components at the top-level; going down one level in component U0 shows 2 instances of meta_harden, one instance of uart_rx, and one instance of led_ctl.

2. To see where the uart_rx_i0 gets generated, right-click on the uart_rx_i0 instance and select *Go To Source* and see that line 84 in the source code is generating it.
3. Double-click on the uart_rx_i0 instance in the schematic diagram to see the underlying components.



Lower level components of the uart_rx_i0 module

4. Click on **Flow Navigator > RTL Analysis > Open Elaborated Design > Report Noise**.
5. Click **OK** to generate the report named **ssn_1**.
6. View the ssn_1 report and observe the unplaced ports, Summary, and I/O Bank Details are highlighted in red because the pin assignments were not done. Note that only output pins are reported as the noise analysis is done on the output pins.

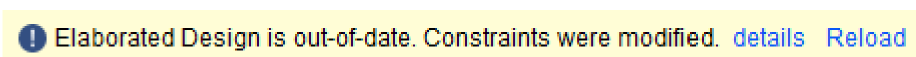


Name	Port	I/O Std	Vcco	Slew	Drive Strength (...	Off-Chip Termina...	Remaining Margin ...	Notes
I/O Bank 0 (0)								
I/O Bank 13 (0)								
I/O Bank 34 (0)								
I/O Bank 35 (0)								
Unplaced Ports (8)								
led_pins[0]	led_pins[0]	LVCMOS18	1.80	SLOW	12	FP_VTT_50		CRITICAL WARNING - Unplaced port
led_pins[1]	led_pins[1]	LVCMOS18	1.80	SLOW	12	FP_VTT_50		CRITICAL WARNING - Unplaced port
led_pins[2]	led_pins[2]	LVCMOS18	1.80	SLOW	12	FP_VTT_50		CRITICAL WARNING - Unplaced port
led_pins[3]	led_pins[3]	LVCMOS18	1.80	SLOW	12	FP_VTT_50		CRITICAL WARNING - Unplaced port
led_pins[4]	led_pins[4]	LVCMOS18	1.80	SLOW	12	FP_VTT_50		CRITICAL WARNING - Unplaced port
led_pins[5]	led_pins[5]	LVCMOS18	1.80	SLOW	12	FP_VTT_50		CRITICAL WARNING - Unplaced port
led_pins[6]	led_pins[6]	LVCMOS18	1.80	SLOW	12	FP_VTT_50		CRITICAL WARNING - Unplaced port
led_pins[7]	led_pins[7]	LVCMOS18	1.80	SLOW	12	FP_VTT_50		CRITICAL WARNING - Unplaced port

Noise report

7. Click on **Add Sources** under the *Project Navigator*, select *Add or create constraints* option and click **Next**.
8. Click on the **Blue Plus** button, then the **Add Files…** button and browse to the **<2018_2_zynq_sources>\lab2** directory, select the **uart_led_pins_pynq.xdc** file, click **OK**, and then click **Finish** to add the pins location constraints.

   Notice that the sources are modified and the tools detect it, showing a warning status bar to re-load the design.



9. Click on the **Reload** link. The constraints will be processed.
10. Click on **Report Noise** and click **OK** to generate the report named **ssn_1**. Observe that this time it does not show any errors (no red).

Synthesize the Design

Synthesize the design with the Vivado synthesis tool and analyze the Project Summary output.

1. Click on **Flow Navigator > SYNTHESIS > Run Synthesis**.

   Click **Save** if the Save Project dialog box is displayed.

   The synthesis process will be run on the uart_top.v and all its hierarchical files. When the process is completed a *Synthesis Completed* dialog box with three options will be displayed.

2. Select the *Open Synthesized Design* option and click **OK** as we want to look at the synthesis output.

   Click **Yes** to close the elaborated design if the dialog box is displayed.

3. Select the **Project Summary** tab

   If you don't see the Project Summary tab then select **Layout > Default Layout,** or click the **Project Summary** icon Σ.

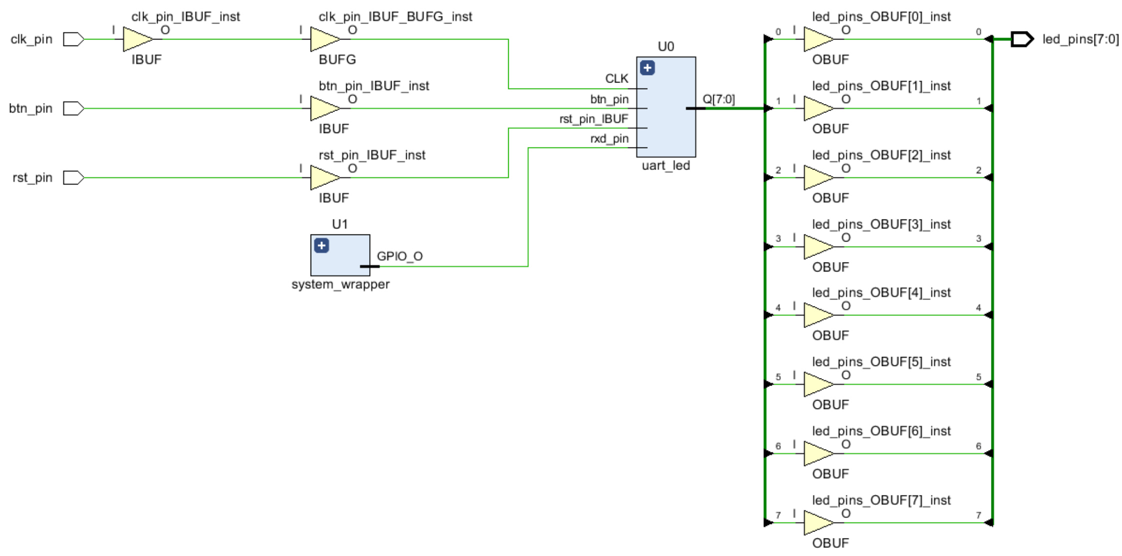
4. Click on the **Table** tab in the **Project Summary** tab and fill out the following information.

Question 1

Look through the table and find the number used of each of the following:

Resource	Number
FF	_
LUT	_
I/O	_
BUFG	_

5. Click on **Flow Navigator > SYNTHESIS > Open Synthesized Design > Schematic** to view the synthesized design in a schematic view.



Synthesized design's schematic view

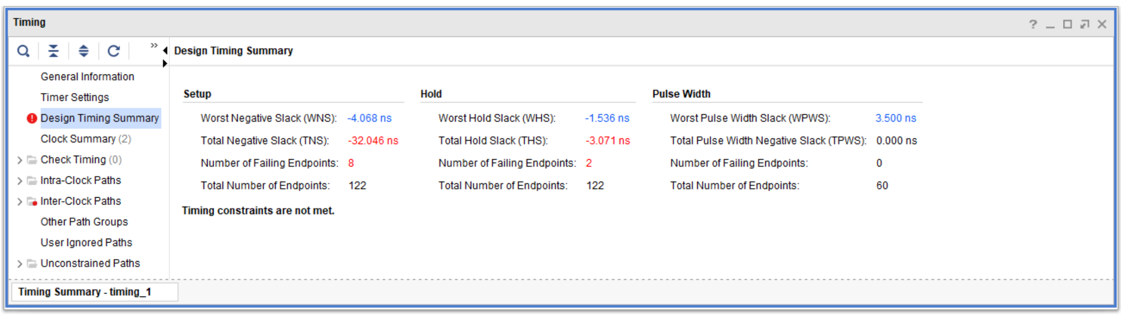
Notice that IBUF and OBUF are automatically instantiated (added) to the design as the input and output are buffered. There are still four lower level modules instantiated.

6. Double-click on **U0** and **uart_rx_i0** instances in the schematic view to see the underlying instances.
7. Select the **uart_baud_gen_rx_i0** instance, right-click, and select *Go To Source*.

   Notice that line 86 is highlighted. Also notice that the CLOCK_RATE and BAUD_RATE parameters are passed to the module being called.
8. Double-click on the **meta_harden_rxd_i0** instance to see how the synchronization circuit is being implemented using two FFs. This synchronization is necessary to reduce the likelihood of meta-stability.
9. Click on the () in the schematic view to go back to its parent block.

Analyze the timing report.

1. Click on **Flow Navigator > SYNTHESIS > Open Synthesized Design > Report Timing Summary**.
2. Click **OK** to generate the Timing_1 report.

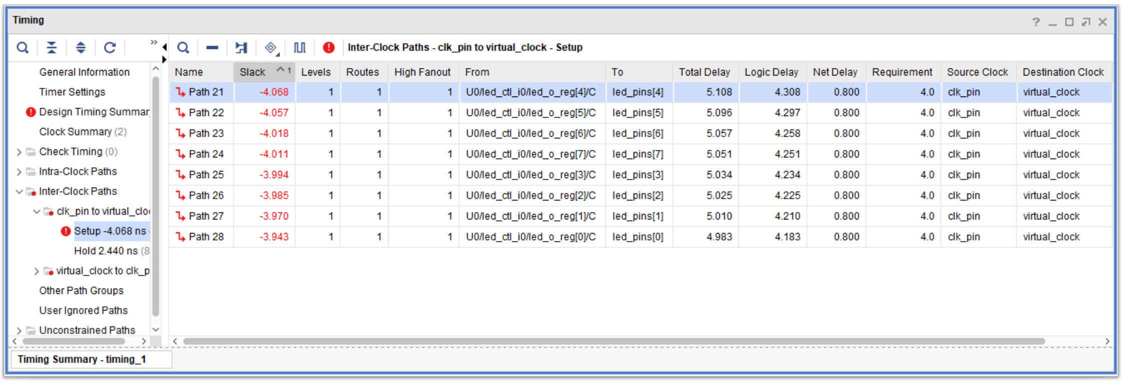


Timing report for the PYNQ

Notice that the Design Timing Summary and Inter-Clock Paths entry in the left pane is highlighted in red indicating timing violations. In the right pane, the information is grouped in Setup, Hold, and Width columns.

Under the Setup column Worst Negative Slack (WNS) is linked indicating that clicking on it can give us insight on how the failing path has formed. The Total Negative Slack (TNS) is highlighted in red indicating the total amount of violations in the design and the Number of Failing Endpoints indicate total number of failing paths.

3. Click on the WNS link and see the 8 failing paths.



The 8 failing paths for the PYNQ

Double-click on the Path 25 to see how the path is made.

Path 25 - timing_1

Summary

Name	Path 25
Slack	-3.994ns
Source	U0/led_ctl_i0/led_o_reg[3]/C (rising edge-triggered cell FDRE clocked by clk_pin {rise@0.000ns fall@4.000ns period=8.000ns})
Destination	led_pins[3] (output port clocked by virtual_clock {rise@0.000ns fall@6.000ns period=12.000ns})
Path Group	virtual_clock
Path Type	Max at Slow Process Corner
Requirement	4.000ns (virtual_clock rise@12.000ns - clk_pin rise@8.000ns)
Data Path Delay	5.034ns (logic 4.234ns (84.113%) route 0.800ns (15.887%))
Logic Levels	1 (OBUF=1)
Output Delay	0.000ns
Clock Path Skew	-2.935ns
Clock Uncertainty	0.025ns
Clock Dom...Crossing	Inter clock paths are considered valid unless explicitly excluded by timing constraints such as set_clock_groups or set_false_path.

Source Clock Path

Delay Type	Incr (ns)	Path (...	Locati...	Netlist Resource(s)
(clock clk_pin rise edge)	(r) 8.000	8.000		
	(r) 0.000	8.000	Sit...16	clk_pin
net (fo=0)	0.000	8.000		clk_pin
			Sit...16	clk_pin_IBUF_inst/I
IBUF (Prop_ibuf_I_O)	(r) 1.451	9.451	Sit...16	clk_pin_IBUF_inst/O
net (fo=1, unplaced)	0.800	10.250		clk_pin_IBUF
				clk_pin_IBUF_BUFG_inst/I
BUFG (Prop_bufg_I_O)	(r) 0.101	10.351		clk_pin_IBUF_BUFG_inst/O
net (fo=59, unplaced)	0.584	10.935		U0/led_ctl_i0/CLK
FDRE				U0/led_ctl_i0/led_o_reg[3]/C

Data Path

Delay Type	Incr (ns)	Path (...	Locati...	Netlist Resource(s)
FDRE (Prop_fdre_C_Q)	(r) 0.478	11.413		U0/led_ctl_i0/led_o_reg[3]/Q
net (fo=1, unplaced)	0.800	12.213		led_pins_OBUF[3]
			Sit...14	led_pins_OBUF[3]_inst/I
OBUF (Prop_obuf_I_O)	(r) 3.756	15.969	Sit...14	led_pins_OBUF[3]_inst/O
net (fo=0)	0.000	15.969		led_pins[3]
			Sit...14	led_pins[3]
Arrival Time		15.969		

Worst failing path for the PYNQ

Note that this is an estimate only. The nets are specified as unplaced and have all been allocated default values (0.584 ns). No actual routing delays are considered.

Generate the utilization and power reports.

1. Click **Flow Navigator > SYNTHESIS > Open Synthesized Design > Report Report Utilization**, and click **OK** to generate the utilization report. Click on **Summary** in the left pane.

Resource	Utilization	Available	Utilization %
LUT	87	53200	0.16
FF	59	106400	0.06
IO	11	125	8.80

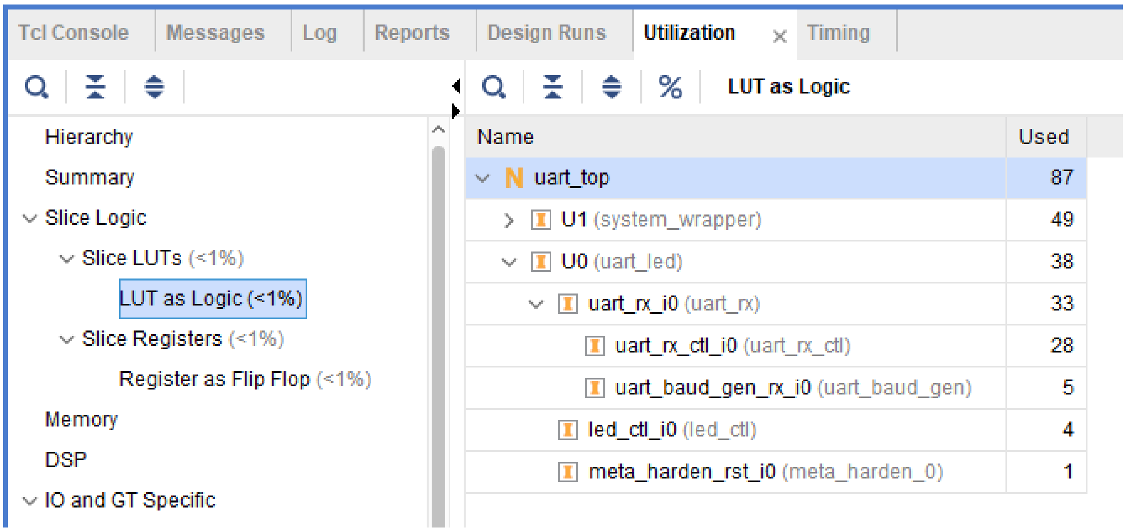
Utilization report for the PYNQ

Question 2

Look through the report and find the number used of each of the following:

Resource	Number
FF:	__
LUT:	__
I/O:	__
BUFG:	__

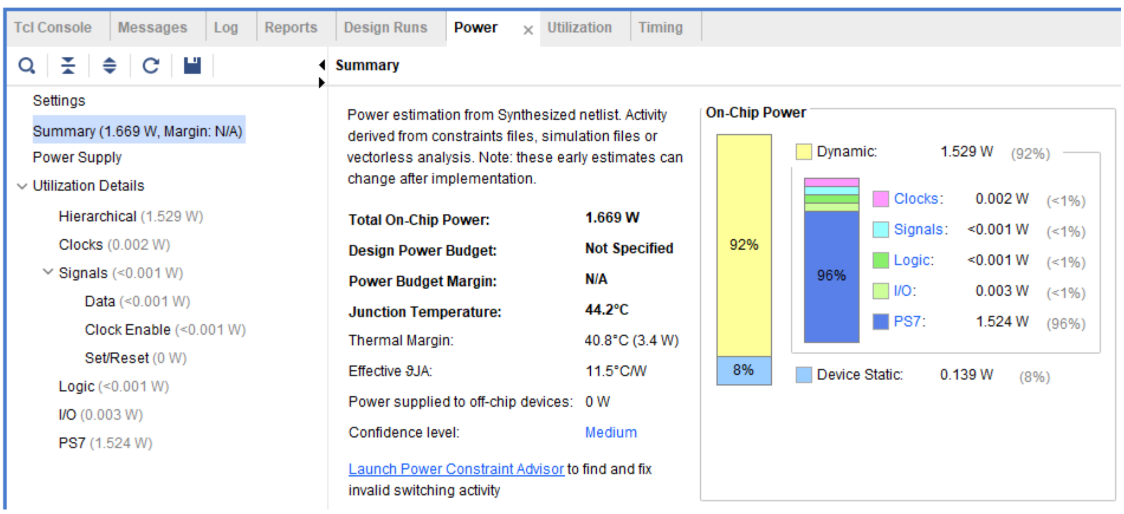
2. Select Slice LUTs entry in the left pane and see the utilization by lower-level instances. You can expand the instances in the right pane to see the complete hierarchy utilization.



Utilization of lower-level modules for the PYNQ

3. Click **Flow Navigator > SYNTHESIS > Open Synthesized Design > Report Power**, and click **OK** to generate the estimated power consumption report using default values.

   Note that this is just an estimate as no simulation run data was provided and no accurate activity rate, or environment information was entered.



Power consumption estimation for the PYNQ

Question 3

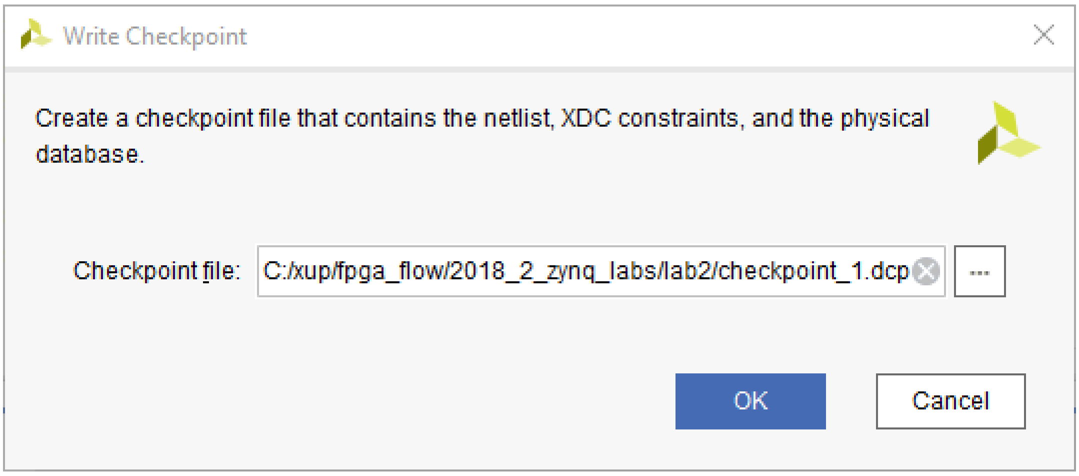
From the power report, find the % power consumption used by each of the following:

Resource	**Number**
Clocks:	_
Signals:	_
Logic:	_
I/O:	_
PS7:	_

You can move the mouse on the boxes which do not show the percentage to see the consumption.

Write the checkpoint to analyze the results without going through the actual synthesis process.

1. Select **File > Checkpoint > Write...** to save the processed design so it can be opened later for further analysis.
2. A dialog box will appear showing the default name of the file in the current project directory.



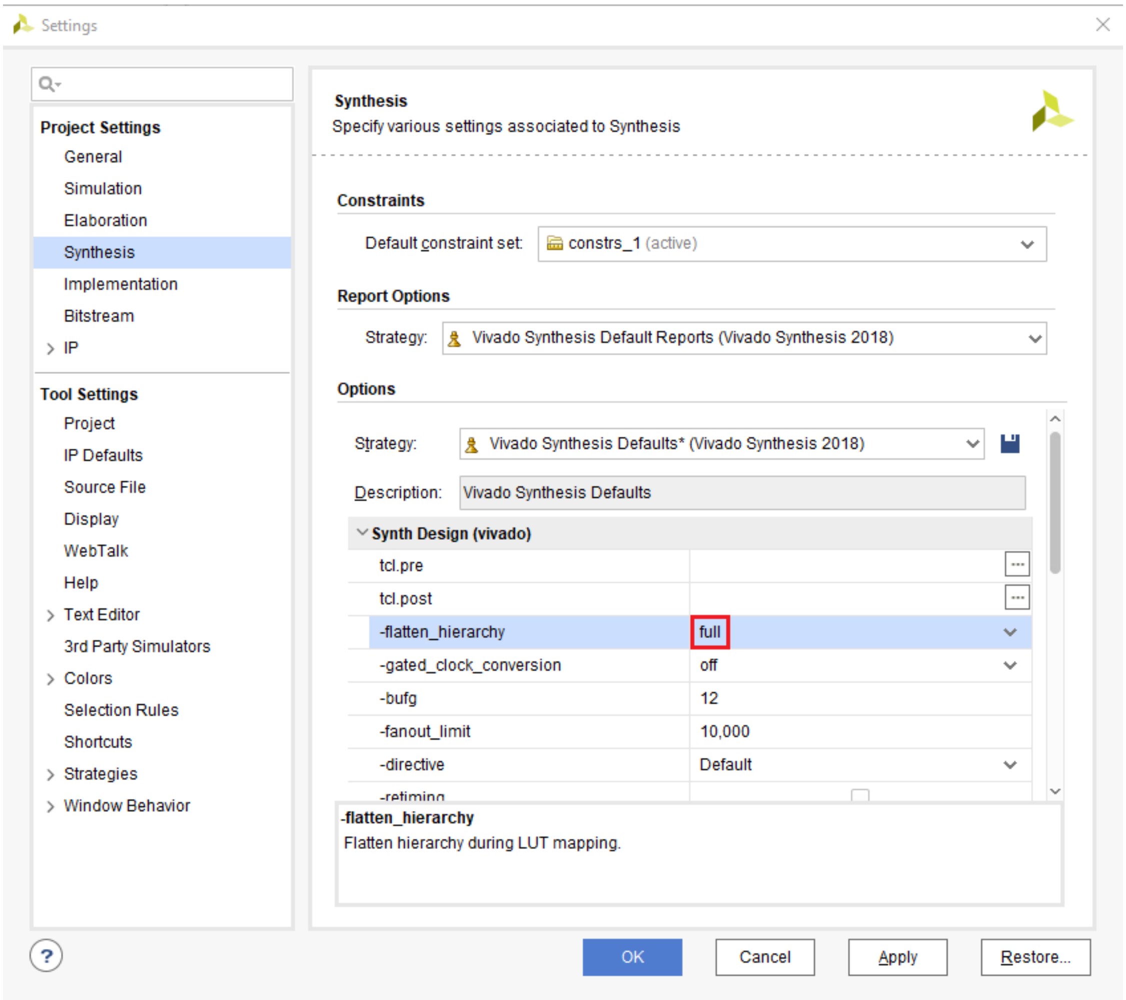
Writing checkpoint

3. Click **OK**.

Change the synthesis settings to flatten the design. Re-synthesize the design and analyze the results

1. Click on the **Settings** under the *Project Manager*, and select **Synthesis**.

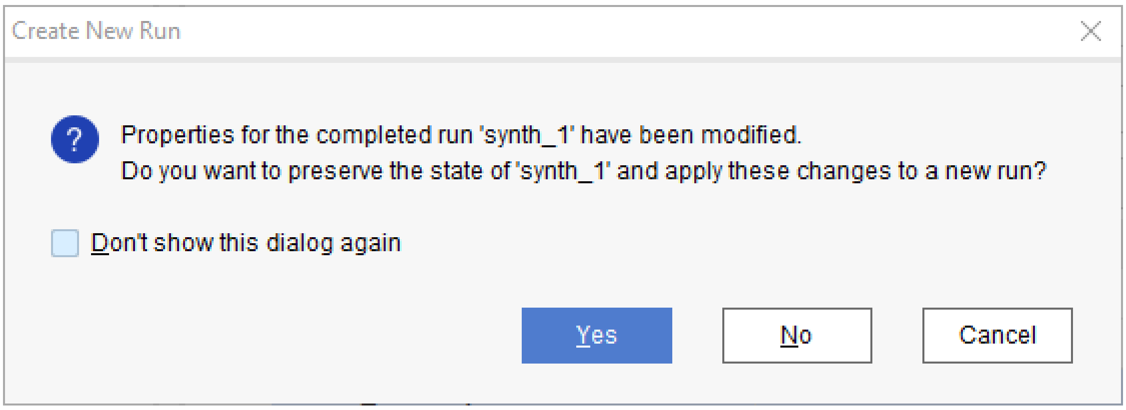
2. Click on the **flatten_hierarchy** drop-down button and select **full** to flatten the design.



Selecting flatten hierarchy option

3. Click **OK**.

4. A Create New Run dialog box will appear asking you whether you want to create a new run since the settings have been changed.



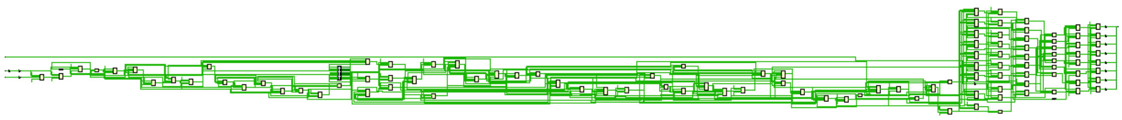
Create New Run dialog box

5. Click **Yes**.

6. Change the name from **synth_2** to **synth_flatten** and click **OK**.
7. Click **Run Synthesis** to synthesize the design.
8. Click **Save**, **OK**, and again **OK** to save the synthesized design and save the constraints.

   The Reload Design dialog box may re-appear. Click **Cancel**.
9. Click **OK** to open the synthesized design when synthesis process is completed.
10. Click on **Flow Navigator > SYNTHESIS > Open Synthesized Design > Schematic** to view the synthesized design in a schematic view.

    Notice that the design is completely flattened.



Flattened design

11. Click on **Report Utilization** and observe that the hierarchical utilization is no longer available. Also note that the number of **Slice Registers** is **59**.

Write the checkpoint to analyze the results without going through the actual synthesis process

1. Select **File > Checkpoint > Write...** to save the processed design so it can be opened later for further analysis.
2. A dialog box will appear showing the default name of the file (checkpoint_2.dcp) in the current project directory.
3. Click **OK**.
4. Close the project by selecting **File > Close Project**.

Read the Checkpoints

Read the previously saved checkpoint (checkpoint_1) in order to analyze the results without going through the actual synthesis process

1. Select **File > Checkpoint > Open...** at the *Getting Started* screen.
2. Browse to **<2018_2_zynq_labs>\lab2** and select **checkpoint_1.**
3. Click **OK**.
4. If the schematic isn't open by default, in the netlist tab, select the **U0**(**uart_led)**, right-click and select **Schematic**.

   You will see the hierarchical blocks. You can double-click on any of the first-level block and see the underlying blocks. You can also select any lower-level block in the netlist tab, right-click and select Schematic to see the corresponding level design.
5. In the netlist tab, select the top-level instance, **U0**(**uart_led)**, right-click and select **Show Hierarchy.**

   You will see how the blocks are hierarchically connected.
6. Select **Reports > Timing > Report Timing Summary** and click **OK** to see the report you saw previously.

7. Select **Reports > Report Utilization…** and click **OK** to see the utilization report you saw previously
8. Select **File > Checkpoint > Open**, browse to **<2018_2_zynq_labs >\lab2** and select **checkpoint_2.**
9. Click **No** to keep the Checkpoint_1 open.

   This will invoke second Vivado GUI.
10. If the schematic isn't open by default, in the netlist tab, select the top-level instance, **uart_top**, right-click and select **Schematic**.

    You will see the flattened design.
11. You can generate the desired reports on this checkpoint as you wish.
12. Close the **Vivado** program by selecting **File > Exit** and click **OK**.

Conclusion

In this lab you applied the timing constraints and synthesized the design. You viewed various post-synthesis reports. You wrote checkpoints and read it back to perform the analysis you were doing during the design flow. You saw the effect of changing synthesis settings.

Answers

1. Look through the table and find the number used of each of the following:

Resource	Number
FF	59
LUT	87
I/O	11
BUFG	1

2. Look through the table and find the number used of each of the following:

Resource	Number
FF	59
LUT	87
I/O	11
BUFG	1

3. From the power report, find the % power consumption used by each of the following:

Resource	Number
Clocks:	<1%
Signals:	<1%
Logic:	<1%
I/O:	<1%
PS7:	<96%